

QUIZ 2

MA 201 Mathematics III

Department of Mathematics, IIT Guwahati

Date: November 3, 2025

Time: 7:55–8:50 AM

Total Marks: 15

1 Solutions:

1. Find the partial differential equation corresponding to the integral surface $u = u(x, y)$ which is orthogonal to the surface $x^2 + y^2 + (u - 1)^2 = 4$. [2 Marks]

Solution:

Let us write $F(x, y, u) = u(x, y) - u = 0$ giving us $\nabla F = \langle u_x, u_y, -1 \rangle$

and $G(x, y, u) = x^2 + y^2 + (u - 1)^2 - 4 = 0$ giving us $\nabla G = \langle 2x, 2y, 2(u - 1) \rangle$.

..... 1 MARK

For the surfaces to be orthogonal, we must have

$$\nabla F \cdot \nabla G = 0$$

$$\Rightarrow \langle u_x, u_y, -1 \rangle \cdot \langle 2x, 2y, 2(u - 1) \rangle = 0$$

$$\Rightarrow xu_x + yu_y = u - 1,$$

which is the required PDE.

..... 1 MARK

[Note regarding a mistake committed by many:

If any one calculated ∇G as

$$G(x, y, u) = x^2 + y^2 + (u - 1)^2 - 4 = 0 \quad \Rightarrow \quad \nabla G = \langle 2x, 2y, 2(u - 1) \rangle,$$

and directly wrote the characteristic equations (in Lagrange form) by

$$\frac{dx}{G_x} = \frac{dy}{G_y} = \frac{du}{G_u}, \quad (1.1)$$

and then computed some integral surface say $H(\phi, \psi) = 0$ and the PDE is obtained by eliminating the unknown function H , then one deserves **0.5 mark** only.

Reason: One cannot write the form (1.1) without knowing the PDE itself!]

2. Obtain the integral surface corresponding to the partial differential equation $2y(u-3)u_x + (2x-u)u_y = y(2x-3)$ and that passes through the circle $x^2 + y^2 = 2x$, $u = 0$. [4 Marks]

Solution:

The auxiliary equations are

$$\frac{dx}{2y(u-3)} = \frac{dy}{2x-u} = \frac{du}{y(2x-3)}$$

which can be written as

$$\frac{y(2x-3)dx}{2y(u-3)} = \frac{y(2x-3)dy}{2x-u} = du.$$

..... 1 MARK

Now the first and the third fractions give us

$$\frac{(2x-3)dx}{2(u-3)} = du,$$

which upon integration yields

$$x^2 - u^2 - 3x + 6u = C_1.$$

..... 1 MARK

Next, we can write

$$\frac{dx + 2ydy}{-6y + 4xy} = \frac{2du}{4xy - 6y},$$

which gives

$$x + y^2 - 2u = C_2.$$

..... 1 MARK

Using the given condition that the surface must pass through $x^2 + y^2 = 2x$, $u = 0$ gives us

$$C_1 + C_2 = 0.$$

Therefore, the required surface is given by

$$x^2 + y^2 - u^2 - 2x + 4u = 0.$$

..... 1 MARK

3. Examine if the partial differential equation $u_x = cu(x, y)$, $c \in \mathbb{R}$ subject to the condition $u(x, 0) = \cos x$ has any solution. [2 Marks]

Solution:

The given problem is $u_x = cu(x, y)$, $c \in \mathbb{R}$, $u(x, 0) = \cos x$.

Although $u = u(x, y)$, the absence of any derivative with respect to y makes it fine to consider this as an ODE for u in x . Separating the variables u and x , we get the solution as

$$u(x, y) = f(y)e^{cx},$$

where $f(y)$ is an arbitrary function.

..... 1 MARK

Using the given condition $u(x, 0) = \cos x$, we get $\cos x = f(0)e^{cx}$. It is obvious that there is no such $f(y)$ which satisfies this relation.

Hence, the problem does not admit any solution.

..... 1 MARK

[**Note regarding a mistake committed by many:**

Some students used the method of characteristics to determine the characteristic curves of the form (say)

$$x(s, t) = s + t \text{ and } y(s, t) = 0. \quad (1.2)$$

They either computed the Jacobian $\frac{\partial(x, y)}{\partial(s, t)} = 0$ or simply inferred from the system (1.2) that s and t cannot be determined uniquely and hence the given initial value problem (IVP) has no solution. Then, one deserves **1 mark** only.

Reason: Even if Jacobian $\frac{\partial(x, y)}{\partial(s, t)} = 0$ or s and t cannot be expressed uniquely in terms of x and y , it can also result in infinitely many solutions.

Those who have written that the given IVP will have infinitely many solutions even after computing the Jacobian to be zero shall **not be awarded** any marks.]

4. Consider the partial differential equation $2(x-y)u_{xx} + 4(x-y)u_{xy} + 4xu_{yy} + x^2u_x - y^2u_y + 9u - 1 = 0$. Classify it (that is, whether hyperbolic/parabolic/elliptic) on the hyperbola $x^2 - y^2 = 9$, the parabola $y = x^2 + 1$ and on the line $x + y = 0$. **[3 Marks]**

Solution:

Comparing the coefficients with the standard second-order PDE for u in two independent variables x, y , we identify

$A = 2(x-y)$, $B = 4(x-y)$, $C = 4x$. Hence,

$$\begin{aligned} B^2 - 4AC &= 16(x-y)^2 - 32(x-y) \\ &= -16(x-y)(x+y) \\ &= -16(x^2 - y^2). \end{aligned}$$

(i) On the hyperbola $x^2 - y^2 = 9$, we have $B^2 - 4AC = (-16) \times 9 = -144 < 0$.

Hence, the PDE is elliptic on $x^2 - y^2 = 9$.

(ii) On the parabola $y = x^2 + 1$, we have $B^2 - 4AC = (-16)(y - 1 - y^2) = 16(y^2 + 1 - y) > 0$.

Hence, the PDE is hyperbolic on $y = x^2 + 1$.

(iii) On the line $x + y = 0$, we have $B^2 - 4AC = -16 \times 0 = 0$.

Hence, the PDE is parabolic on $x + y = 0$.

Marking scheme:

If one obtains only the expression of $B^2 - AC$ correctly and gets none of (i), (ii) and (iii) correctly, he/she gets 1 Mark.

If one obtains the expression of $B^2 - AC$ correctly and gets only one of (i), (ii) and (iii) correctly, he/she gets 1.5 Marks.

If one obtains the expression of $B^2 - AC$ correctly and gets only two of (i), (ii) and (iii) correctly, he/she gets 2 Marks.

5. Write D'Alembert's solution of the one-dimensional wave equation $u_{tt} = c^2 u_{xx}$, $-\infty < x < \infty$ with the initial conditions $u(x, 0) = 0$, $u_t(x, 0) = \cos x$. **[1 Mark]**

Solution:

Since the initial displacement is zero, the solution is

$$u(x, t) = \frac{1}{2c} \int_{x-ct}^{x+ct} \cos s ds = \frac{1}{2c} \left[\sin(x+ct) - \sin(x-ct) \right].$$

..... **1 MARK**

6. Find the Fourier series expansion for the function $f(x)$ as given below:

$$f(x) = \begin{cases} -\pi/2, & -\pi < x < 0, \\ \pi/2, & 0 < x < \pi. \end{cases}$$

Using this Fourier series, find the sum of the infinite series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

[2+1 Marks]

Solution:

It is an odd function.

$$\text{Hence } A_n = 0, \quad B_n = \frac{2}{\pi} \int_0^\pi \frac{\pi}{2} \sin nx dx = \frac{(1 - \cos n\pi)}{n} = \begin{cases} \frac{2}{n}, & \text{for } n \text{ odd,} \\ 0, & \text{for } n \text{ even.} \end{cases}$$

$$\text{Hence, } f(x) \approx 2 \sum_{n=1}^{\infty} \frac{\sin((2n-1)x)}{2n-1}.$$

.....2 MARKS

Since $x = \pi/2$ is a point of continuity, we can write $f(x) = 2 \sum_{n=1}^{\infty} \frac{\sin((2n-1)x)}{2n-1}$ at $x = \pi/2$. Hence,

$$\pi/2 = 2 \sum_{n=1}^{\infty} \left[\frac{\sin((2n-1)\pi/2)}{2n-1} \right].$$

This implies

$$\pi/2 = 2 \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right].$$

It gives

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \pi/4.$$

..... 1 MARK